Vortex86EX-A9123

Layout Guideline

Ver. 1.0

Modify	Date (mm/dd/yy)	Version	Description
	2012/11/20	0.1	Initial Release
Joe Y.	2012/11/23	0.2	1. Add crystal recommend 2. Modify the stack up for 6-layer and 4-layer PCB 3. Modify DDR layout guide for 2 rank 4. Add power up/down sequence 5. Remover external clock generator for SATA and PCIe
Joe Y.	2012/12/21	0.3	1. Modify damping resistor for 25MHz crystal and crystal specification (change to 200 uW typical)
Joe Y.	2013/01/08	0.4	1. Power pins connection of internal regulator controller if internal regulator no need.
Joe Y.	2013/4/18	0.5	1. Add DDR Vref circuit description 2. Modify the trace length constrain for 1 or 2 DDR3 device
Joe Y.	2013/4/19	0.6	1. Modify the DDR3 interface reference voltage of A9123
Joe Y.	2013/7/19	0.7	1. Remove the constrain for bootstrap on DDR interface
Joe Y.	2013/7/22	0.8	1. Add internal regulator guide 2. Add PCIe and SATA guideline when PCIe and SATA interface unused 3. Add high speed signal guideline when high speed signal connect to other board by board to board connector.
Joe Y.	2013/11/01	0.9	1. Add ISA Bus interface guideline 2. Add RTC guideline when RTC interface unused 3. Add internal PCIe PLL specification and measured result 4. PCIe external clock generator guideline 5. Add USB port power source notice 6. Add DDR3 support list
Joe Y.	2014/5/27	1.0	1. Add Ethernet protection scheme for connect to POE device 2. Add ADC port protect circuit

Confidential

Overview

1.1. General Description

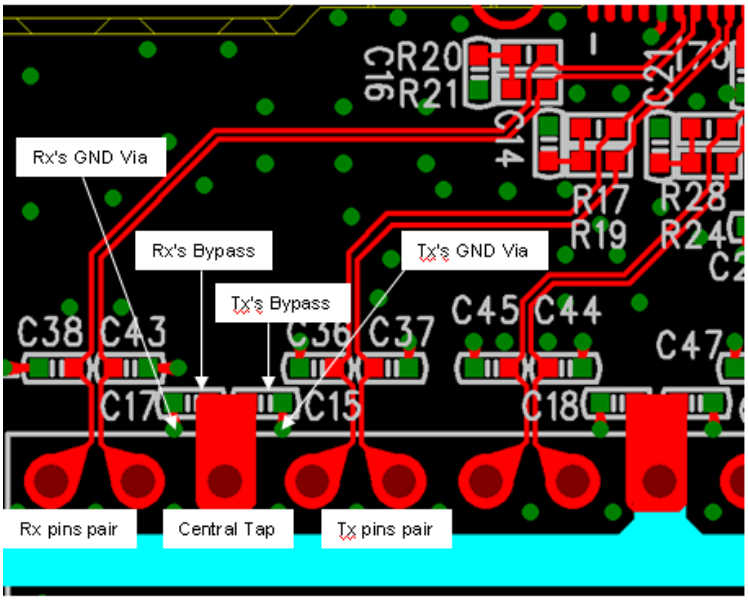
The following document provides the technical data that describe layout guideline of Vortex86EX (so called A9123). The main purpose of this document was to provide A9123 with high quality signal for system reliable operation.

1.2. Ethernet PHY Layout Guideline

Place termination resistor near IC side. Route differential pair of Ethernet PHY with 4.5mil traces width and 5mil space between TXP (RXP) and TXN (RXN), keep minimum 30 mils space for other traces to reduce crosstalk and EMI as possible. Keep the analog signals away from digital signals. Place resistor of ISET near A9123 at least with trace width of 20mil. Place Transform near RJ45 connector and place termination resistor near IC side.

Pin U1 (AVSS_EPHYTX18) is the ground pin of transmitter; it can't share via with other analog ground pins. It can't through bead connect to system ground. Strong suggestion connection to clear system ground directly.

It is importance for the layout of transformer's central tap. Due to our PHY support auto MDI/MDIX. Add the bypass capacitor of central tap for transmit path and receiver path with ground via near transmit and receiver pins of transformer respectively. The figure is as follows:

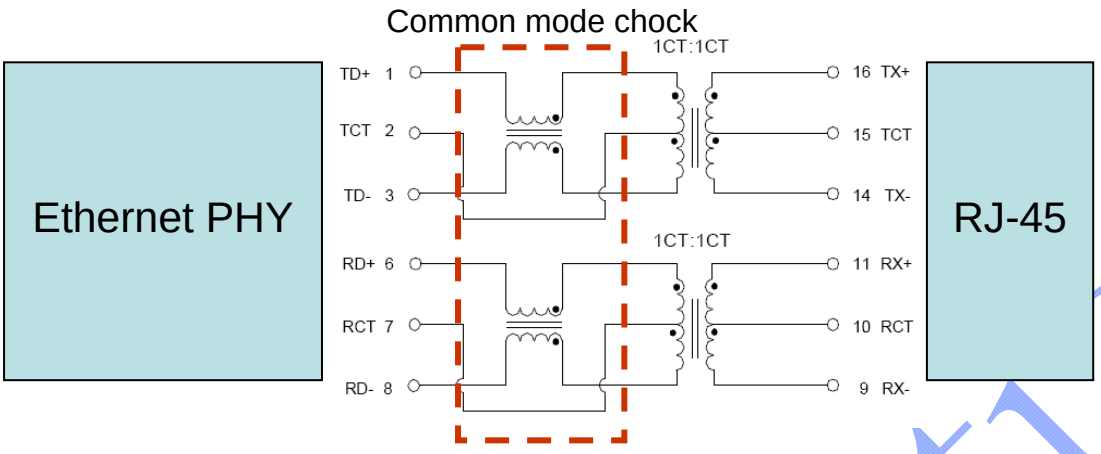


Ethernet's Layout guideline

Confidential

The LAN transformer selection guide is as follow:

Select the LAN transformer with common chock on the PHY side to enhance EMI's performance.

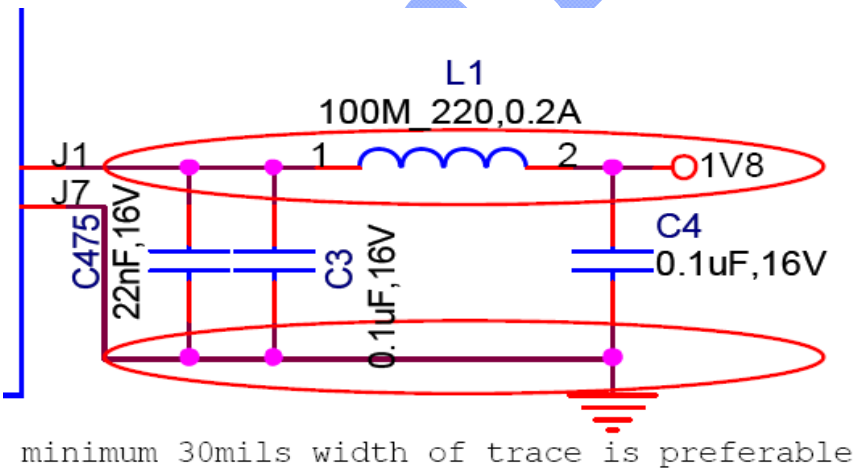


1.3. USB PHY Layout Guideline

Route differential pair of USB PHY with 5mil traces width and 5mil space between DP and DN, keep minimum 30 mils space for other traces to reduce crosstalk and EMI as possible. Keep the analog signals away from digital signals. Place the resistor of ISET near A9123 at least with trace width of 20mil. Place common chock near connector if present.

1.4. LL layout Guideline

A clear power supply is importance for PLL in high performance CPU. DMP recommend a filter technology to reduce jitter as follows:



Rely on A9123 IC side the second layer must be the ground layer.

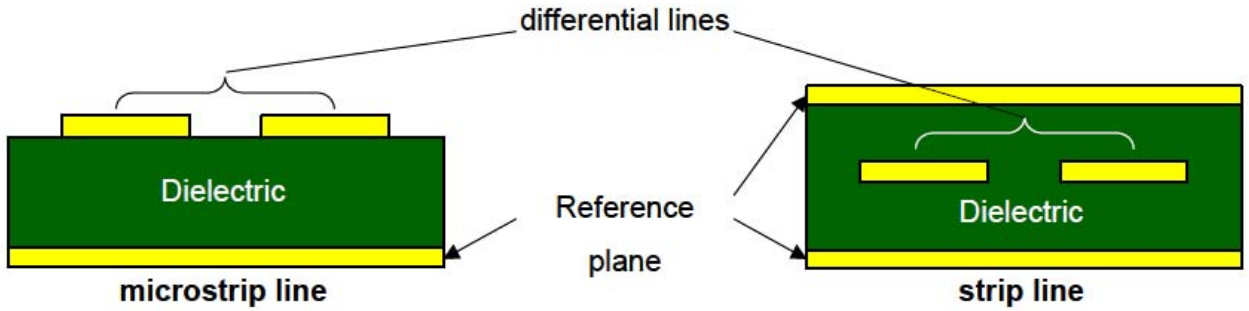
Confidential

1.5. SD Card Interface Layout Guideline

Route traces length of SD Card Interface within 1000mil.

1.6. SATA and PCI EXPRESS Interface Layout Guideline

1. The characteristics impedance of transmission lines must be 100 ohm +/- 10% for differential mode and maximum 35 ohm for common mode of S-ATA and PCIe connection in principle. For appropriate characteristics impedance, the strip line or microstrip line structure must be used. Selection of strip line or microstrip line should be examined on a case-by-case basis according to package (socket), pin count and macro count.



Differential lines structures

2. SATA RX and SATA PLL can common power but, SATA TX power is absolutely not in front of two shared power.

1.7. DDR3 Layout Guideline of 2 Ranks for x8 DDR3 devices

According to the below-mentioned guideline, some constraints were appended as follows:

1. Control the impedance of trace between 55 and 60 ohm.
2. Control the trace length of DQS/DQS# between 1700 and 2000 mil.
3. Control the trace length of DQ/DQM is shorter than DQS/DQS# between 0 and 700 mil.
4. The damping resistor is 10 ohm for data bus.
5. The damping resistor is 0 ohm for address, command and clock bus.
6. The termination resistor is 100 ohm for clock bus and the location is at the first branch for A9123 to DDR3.
7. Control the trace length of CLKP/CLKN is longer than DQS/DQS# between 0 and 300 mil

Confidential

1.8. DDR3 Interface layout Guideline for 1 or 2 DDR3 device(s)

More attention must be paid to high speed rule for board layout and bus loading to make sure that short clock cycle time can be met without noise, ringing, crosstalk power or ground bounce. To provide a good return path and limit above noise on the DDR interface, all system memory signals should reference a single contiguous plane as possible.

The recommendation layout guideline for a DDR device is as follows:

1. In order to attain highest speed as possible. We suggest the signals of DQS/DQS#, DQM and DQ only a loading preference. Please contact with DMP if the signals for DQS/DQS#,DQM and DQ need to load more than one.
2. The DDR clock is differential signal that need a 100 ohm differential terminal resistor near DDR side for a DDR configuration or on split point for two DDRs configuration, and keep the trace length under 0.2 inch from split point to respective DDR. Reserve location of two 0 ohm damping resistors near CPU side for low power application. The two trace width is 4.5 mil and space 5mil to get near 100ohm differential impedance, and keep minimum 15 mils space for other traces to reduce crosstalk as possible. The trace length of two differential signals must be match within 10mil. The category signals were routed within 1.2 inch.
3. Routing each signals of byte lane on same layer with same via count. Control the number of via within 2. The mismatch of trace length must be within 20mil for each signals of byte lane. The high byte lane is DQS1/DQS1#, DQM1 and DQ8~DQ9 and the low byte lane is DQS0/DQS0#, DQM0 and DQ0~DQ7 respective. Place a damping resistor on middle of each signal to trade off the timing of read and write for low power with ODT off. The DQS/DQS# is differential signal that they were routed differential with 4.5mil trace width and 5mil space between DQS and DQS#, keep minimum 15 mils space for other traces to reduce crosstalk as possible. The other signals were routed 5mil trace width and keep minimum 10mil space between signals to reduce crosstalk and control impedance of trace. The high byte lane and low byte lane must be matching within 0.2 inch. The category signals were routed within 1.0 inch. It is importance to match the differential signals of DQS and DQS# within 10mil.
4. Routing signal of command and address with a 10 ohm damping resistor near CPU side. The category signals were routed 5mil trace width with minimum 10mil space between signals to reduce crosstalk and control impedance of trace. The signals of command and address must be matching within 0.1 inch. The category signals were routed within 1.3 inch.
5. To provide a good return path, all system memory signals should reference a single contiguous plane as possible.
6. DDR memory signal trace edges must be at least 4 times (20 mils) trace width from any non-DDR memory signal trace edge.
7. Routing trace near edge of the reference plane should keep minimum 5 times trace width from edge of signal

Confidential

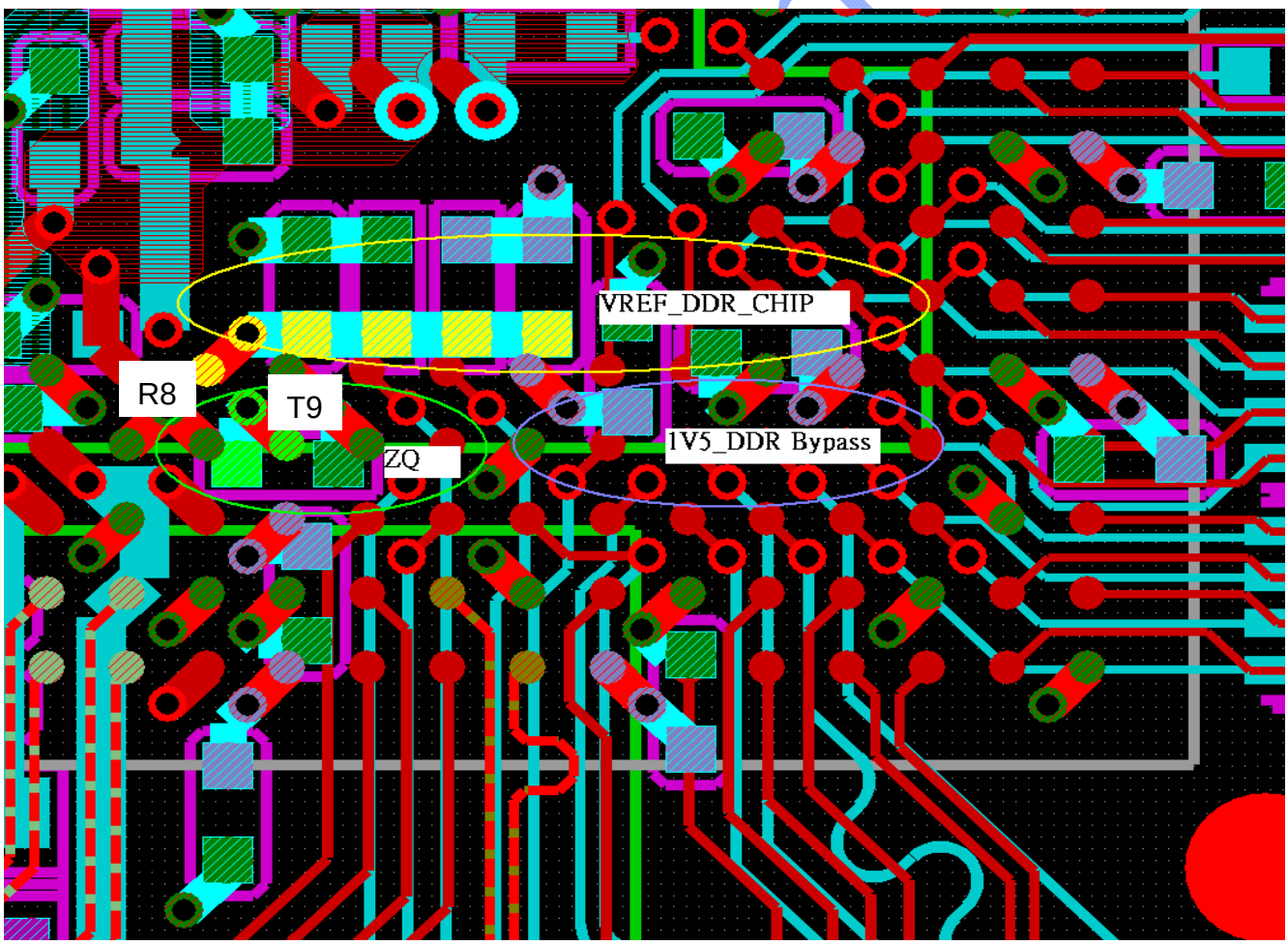
trace to edge of the reference plane.

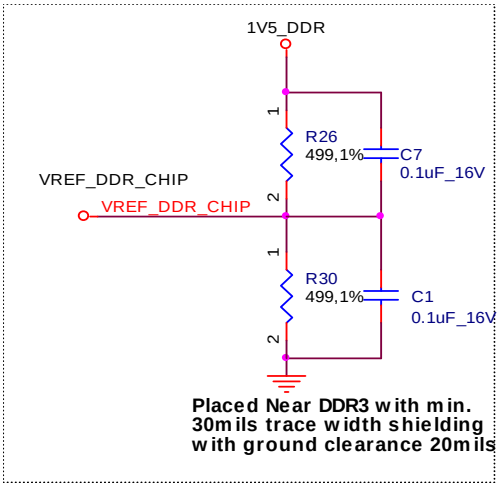
8. The signals of address and command must be following Balanced-T routing rule for the application of two DDR devices.

9. The generation of Vref is critical. It must be follow the variance of VDDQ. We are suggestion to generate the reference from VDDQ by two 499ohm divide resistors with parallel 0.1uf X7R type bypass capacitor for each resistor. Place the divide resistors and bypass capacitor on quiet area. Also place a 0.1uf X7R type bypass capacitor on Vref pin of each device.

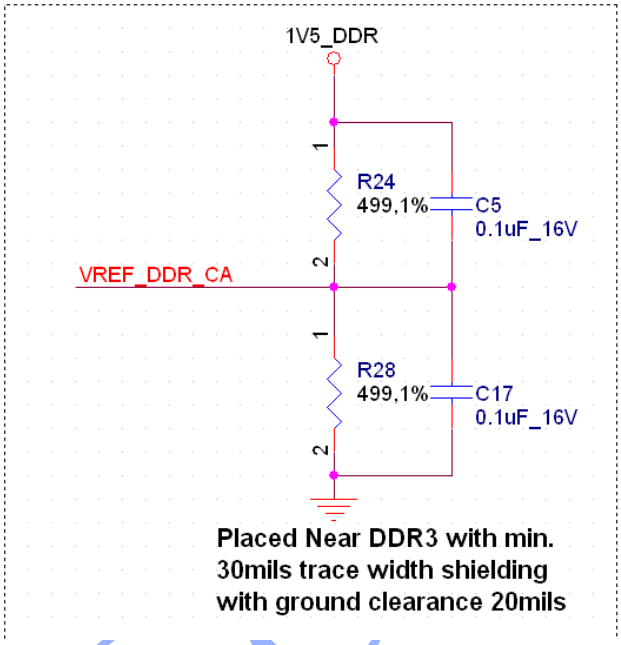
10. The circuit of Vref is as follows:

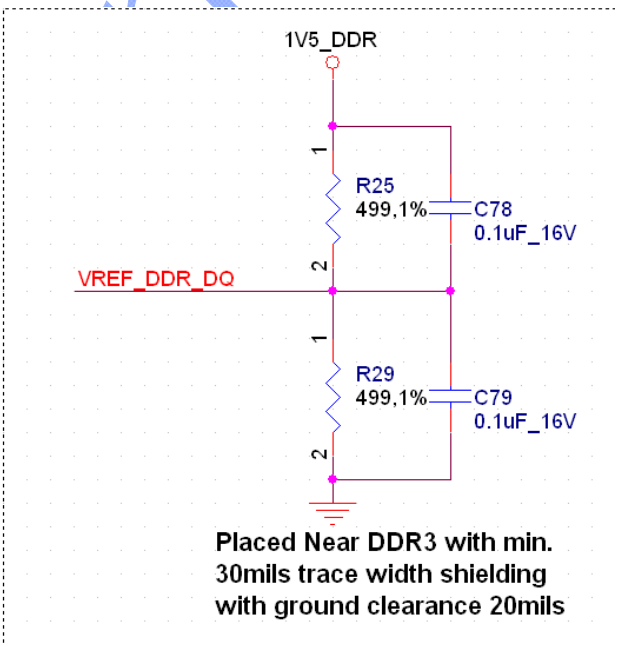
    The DDR3 interface reference voltage of A9123 is critical for power noise. In order to reduce the power noise of reference voltage and follow the DDR3 interface voltage of A9123. Please place the resistor divider and bypass capacitor of reference voltage under the A9123 next to the pin R8 and place the RZQ under the A9123 next to the pin T9. The layout of suggestion is as follows for the DDR3 reference of A9123:





Vref power have two independent reference voltage for DDR3 device , VREF_DDR_DQ provide the reference voltage for DQ, VREF_DDR_CA provide the reference voltage for CA.





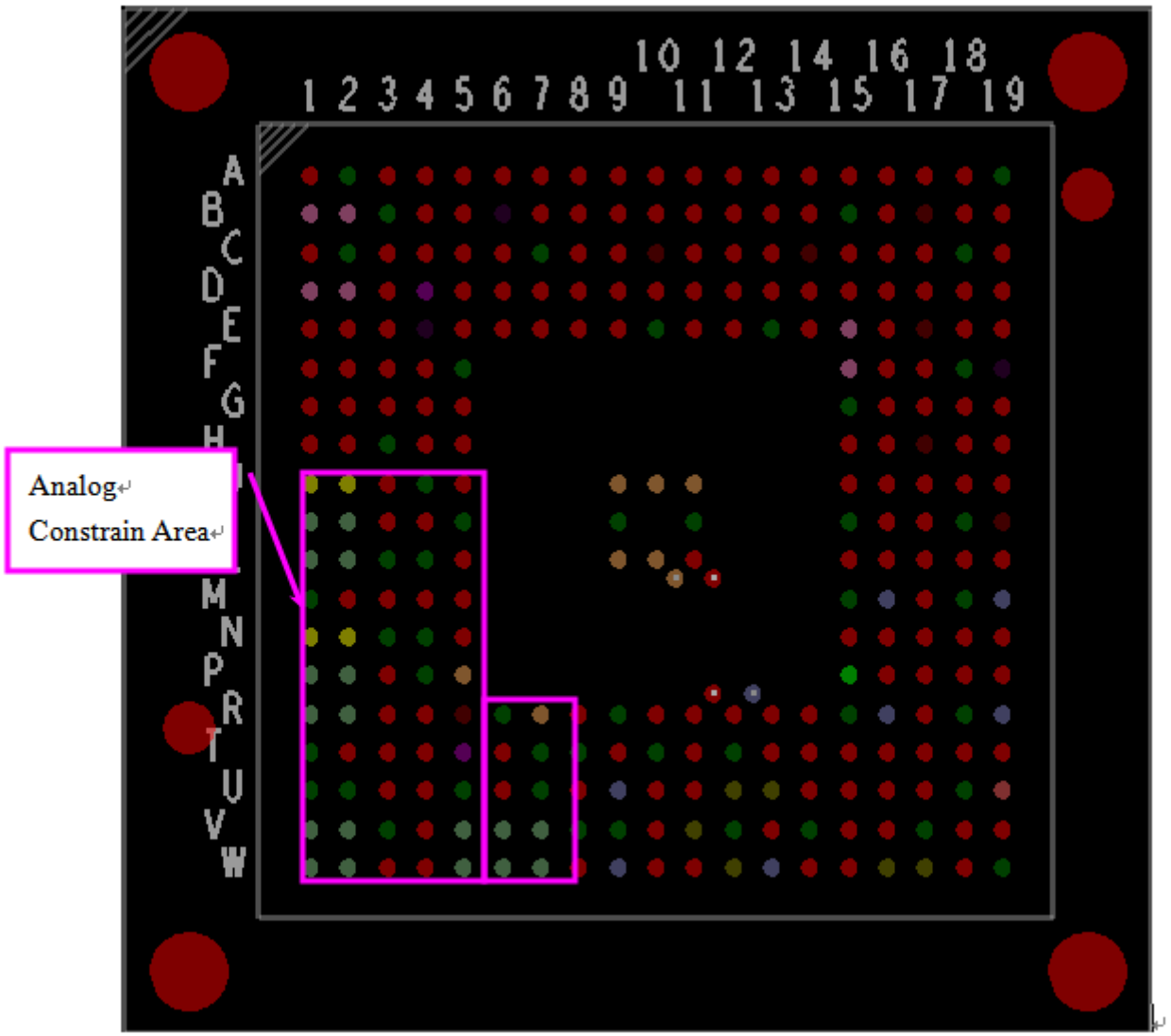
11. Routing Vref as short as possible with minimum 30mil trace width and at least 30mil space from other signal. Routing trace should keep minimum 5 times trace width of next trace between high speed signals to reduce crosstalk.

12. Matching trace length by serpentine line of “S” pattern. Make sure at least 5 times the distance of the trace to reference layer between line segments of “S” that parallel for the application of microstrip.

13. We are suggestion placed a bypass capacitor near vias every 4 to 5 vias for signal transition reference plane.

1.9. Analog area Layout Guideline

Avoid any digital signal cross the area. The constraint area for A9123 is as follows:

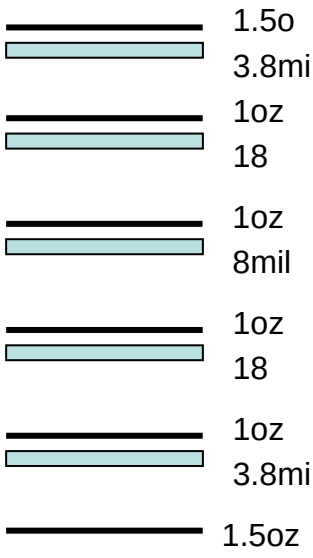


1.10. PCB Stack up

The recommendation PCB stack up is as follows for 4 layers PCB:



The recommendation PCB stack up is as follows for 6 layers PCB:



1.11. Bypass Capacitor scheme

The role of the bypass capacitor is to decouple the power supply bus from the IC. Bypass a power bus from the device internal noise requires capacitor with very low inductance. The objective is to eliminate the effects of the power bus inductance and resistance so that transient currents flowing across the power bus do not cause excessive noise at the power and ground pins of the IC. Therefore, the bypass capacitor should have low effective series resistance (ESR) and series inductance (ESL) while having a large enough capacitance value to supply current to the IC during switching.

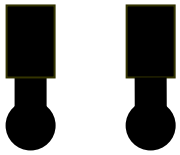
The recommendation bypass scheme for the large surge current is as follows:

1. It is recommended that at least a via per power pin for SDRAM memory interface
2. It is recommended that multiple vias per bypass capacitor to further reduce inductance.
3. It is recommended that X7R or X5R type and 0402 size capacitor for bypass.

Confidential

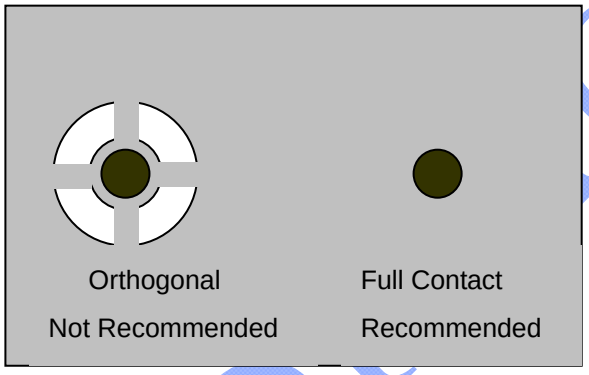
4. It is recommended that a 47uf bulk capacitor for the interface.

5. The land pattern of bypass capacitor is as follows:



1.12. Suggestion Via for DDR3 and A9123

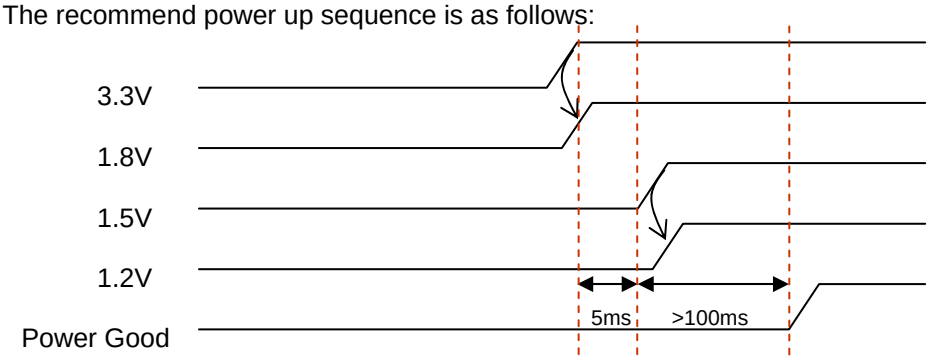
In order to minimum the series inductance, ground and power via should be maximized as possible. We are suggestion on DDR deice that via is at least 18 mil pads with 10mil drilling size. We are also suggestion on A9123 device that via is at least 20 mil pads with 10mil drilling size. The via pattern connect to power/ground plan is as follows:



Confidential

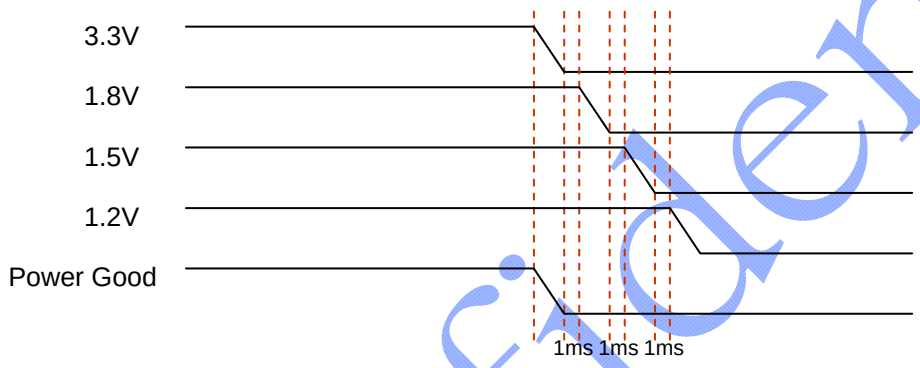
1.13. POWER ON SEQUENCE

The recommend power up sequence is as follows:



1.14. POWER OFF SEQUENCE

The recommended power down sequence is as follows:



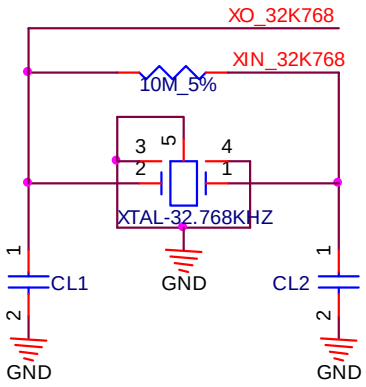
1.15. Crystal recommend TBD for 32.768KHz and 25MHz

The recommend specification of 32.768KHz crystal is as follows:

1. The product must meet RoHS Directives.
2. Aurum Part Number : XF66M400003290
3. Holder Type : FORK-2060
4. Center Frequency : 32.768 KHz
5. Frequency Tolerance (at 25+/-3 deg.C) : +/- 20 ppm max.
6. Frequency Stability : -0.036+/-0.006 ppm/deg.C^2
(over operating temperature)
7. Operating Temperature : -10 ~ 70 deg.C
8. Storage Temperature : -40 ~ 85 deg.C
9. Equivalent resistance(ESR) : 35k Ohm max.
10. Load Capacitance (CL) : 9.0 +/-0.1 pF
11. Shunt capacitance (C0) : 1.7 pF typ.
12. Capacitance ration : 550 typ.
13. Drive Level (DL) : 1.0 uW typ.
14. Insulation resistance : 500M ohm min. at D.C 100V

nfidential

The recommend circuit of 32.768KHz crystal is as follows:



The parasitical capacitance of 5mil PCB trace width on outer layer is about 1.1835pF/cm for 3.8mil P.P reference to shield layer.

The total capacitance of CL1+x*1.1835+3=CL1_total

Where x is trace length crystal pin to Xout pin of A9123.

The total capacitance of CL2+y*1.1835+3=CL2_total

Where y is trace length crystal pin to Xin pin of A9123.

CL=CL1_total*CL2_total/(CL1_total+CL2_total)

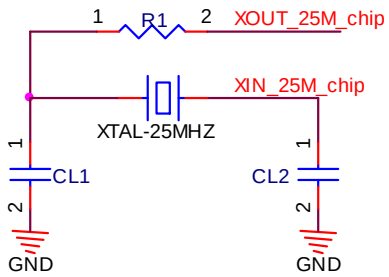
CL is the load capacitance of crystal.

The recommend specification of 25MHz crystal is as follows:

5. Center Frequency (CF)	: 25.000 MHz
6. Oscillation mode / Blank cut mode	: Fundamental / AT-Cut
7. Frequency Tolerance (at 25+/-3 deg.C)	: +/- 30 ppm max.
8. Frequency Stability (over operating temperature)	: +/- 30 ppm max.
9. Operating Temperature	: -40 ~ 85 deg.C
10. Storage Temperature	: a. 25+/-5 deg.C / 45+/-15%R.H. b. -40 ~ 85 deg.C
11. Equivalent series resistance(ESR)	: 25 ohm max.
12. Load Capacitance (CL)	: 12 +/-0.1 pF
13. Shunt capacitance (C0)	: 5.0 pF max.
14. Drive Level (DL)	: 200 uW typ.
15. Aging	: +/-3.0 ppm max.
16. Insulation resistance	: 500M ohm min. at D.C 100V

Confidential

The recommend circuit of 25MHz crystal is as follows:



The R1 is damping resistor to control the drive level under the crystal specification.

R1 of 470ohm is recommended for the 25MHz crystal.

The total capacitance of CL1+x*1.1835=CL1_total

Where x is trace length crystal pin to R1 pin of damping resistor

The total capacitance of CL2+y*1.1835+3=CL2_total

Where y is trace length crystal pin to Xin pin of A9123.

CL=CL1_total*CL2_total/(CL1_total+CL2_total)

CL is the load capacitance of crystal.

1.16. Power pins connection of internal regulator controller if internal regulator no need

Connect power pin (pin U3) of internal regulator controller to 3.3V and to leave unconnected for pins of control and feedback (pin P3, R3, T3 and U4).

1.17. Internal regulator use condition with CPU Frequency under 300MHz for Pins connection of internal regulator controller

1. Pin of voltage control for 1.8V regulator(pin P3) have to connect a BJT component(2SA1577T106R), and Pin of voltage control for 1.2V regulator(pin U4) have to connect a BJT component(2SB1708TL), all this BJT component place near by A9123. Pin of feedback from 1.2V regulator (pin T3) should be 2K/10K ohm divide resistor and near by A9123. Voltage control and feedback for regulator trace width was routed 20mils at least, keep minimum 15 mils space for other traces to reduce crosstalk as possible. Below table is BJT support list:

Vendor	BJT Support List	
	For 1.2V Internal Regulator	For 1.8V Internal Regulator
ROHM	2SB1708TL	2SA1577T106R

Confidential

2. Internal Regulator maximum loading current as follows table:

Maximum Load Current for Internal 1.2V and 1.8V Regulator	
1.2V Internal Regulator(mA)	1.8V Internal Regulator(mA)
1000	300

3. Internal Regulator performance with Loading Transition test result as follows table:

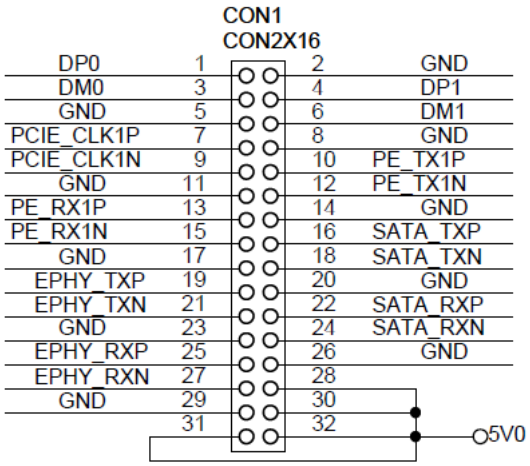
Loading Range	1.2V internal REG ➔Max Loading~1100mA 1.8V internal REG ➔Max Loading~300mA		
REG Voltage(V)	Min	Mean	Max
1.2V	1.139	1.203	1.267
1.8V	1.701	1.771	1.846

1.18. Pins connection of PCIe and SATA interface if no need

1. For power interface of PCIe, connect power pin(pin M4,M3 and T2) to 1.2V, connect power pin(pin N5) to 3.3V For PCIe Tx, Rx and clock interface, to leave unconnected for Tx and Rx pins of P1,P2,R1 and R2.To leave unconnected for clock pins of N1,N2,V6,W6,V7 and W7.
2. For power interface of SATA , connect power pins (pin K3, K4 and M2) to 1.2V,connect power pin(pin L5) to 3.3V, For SATA Tx, Rx and clock interface, to leave unconnected for Tx and Rx pins of K1,K2,L1 and L2. To leave unconnected for clock pins of J1, J2, V5 and W5.

1.19. Pins connection guideline for high speed signal connect to other board by board to board connector

1. For high speed signal connector pin define, like as USB differential pair , PCIe differential pairs, SATA control differential pairs and Ethernet differential pairs all these high speed signal, Below example pin definition is for high speed signals that were recommend to reduce crosstalk:



Confidential

2. For the pin definition of high speed single end signal connector, like as clock signal, next pin should be defining to GND signal for shielding, otherwise, addition one GND signal each 4 single end signals to reduce crosstalk, the pin definition of example connector as follows:

CON2
CON2X20

CLOCK	1		2	Single end1
GND	3		4	Single end2
Single end5	5		6	Single end3
Single end6	7		8	Single end4
Single end7	9		10	GND
Single end8	11		12	Single end9
GND	13		14	Single end10
Single end13	15		16	Single end11
Single end14	17		18	Single end12
Single end15	19		20	GND
Single end16	21		22	Single end17
GND	23		24	Single end18
	25		26	Single end19

1.20. ISA Bus interface layout guideline

For ISA bus signal , recommend trace width is 5 mil and space 5mil to get near 55ohm trace impedance, and keep minimum 15 mils space for other traces to reduce crosstalk as possible. The trace length of data bus signals must be match within 500mil. The trace length of address and control signals must be match within 500mil.The category signals trace length were routed within 3.5 inch.

1.21. Pins connection of RTC interface if no need

For power interface of RTC, connect power pin(pin H4) to 3.3V,RTC power sense pin of J5 connect one 4.7K ohm resistor to 3.3V, to leave unconnected for RTC clock pin of H1 and H2.

1.22. Specification of PCIe internal clock and measured result

1.22.1 As follow table is specification of PCI Express 1.1

	Description	Symbol	Limit	Units
Common Refclk Architecture				
PCIe 1.1	Random Jitter	Rj	4.7	ps pk-pk
	Deterministic Jitter	Dj	41.9	ps pk-pk
	Total Jitter Where Tj = Dj + 14.069 x Rj (for BER 10^{-12})	Tj	108	ps pk-pk

1.22.2 As follows table is measurement of A9123 internal PCIe clock

Confidential

Internal PLL Jitter measurement-A:

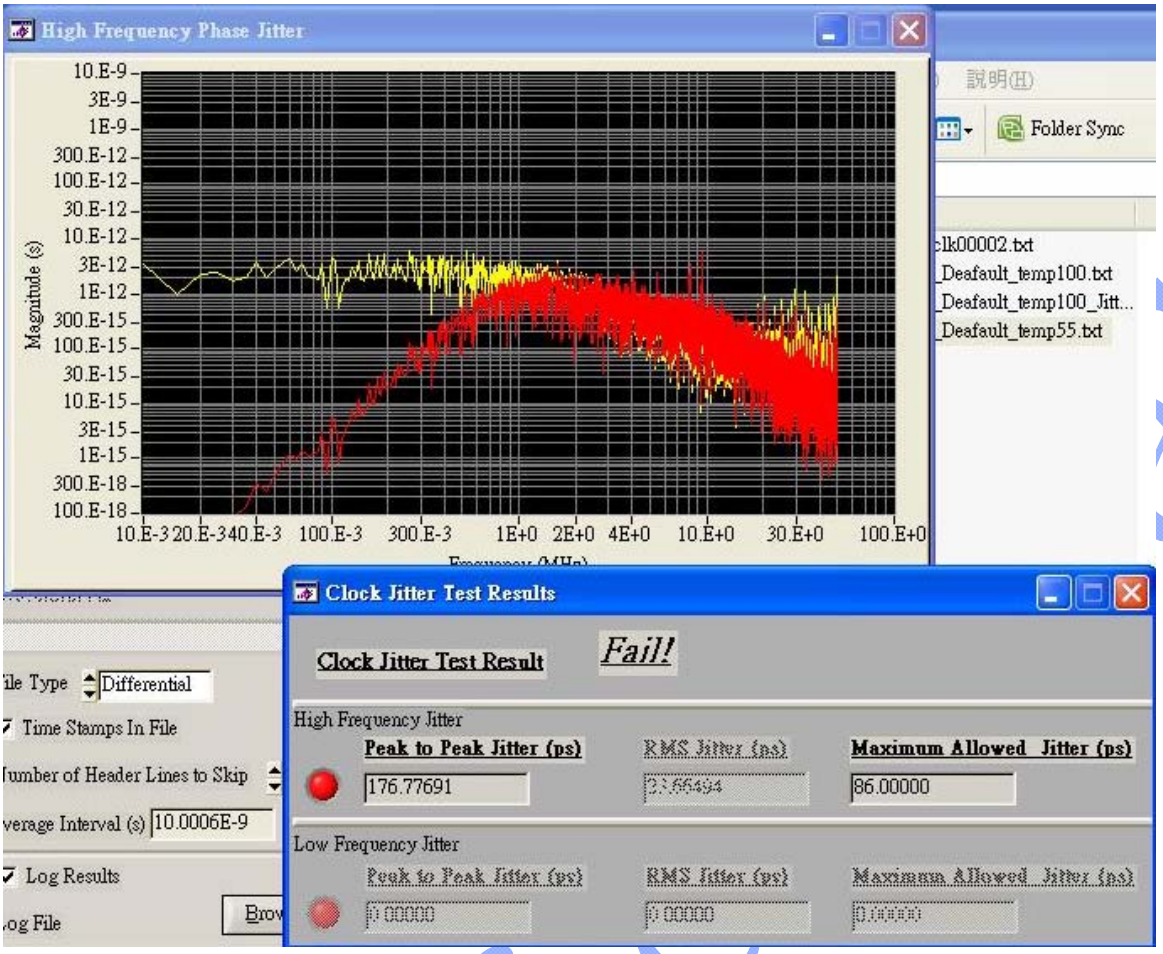
	Internal @55℃	Internal @100℃
Period Jitter (ps) @150M Samples	-54.4~59.9 @P.1	-63.2~61.8 @P.4
Cycle-to-Cycle Jitter (ps) @150M Samples	-78.4~77.5 @P.1	-93.6~93 @P.4
Long Term Jitter (ps) [Delay Time=-10uS] beyond 0.5hr	504 @P.2	490 @P.5
Peak-to-Peak Phase Jitter (ps) by Clock Jitter Tool	176.78 @P.3	168.28 @P.6

Internal PLL Jitter measurement-B:

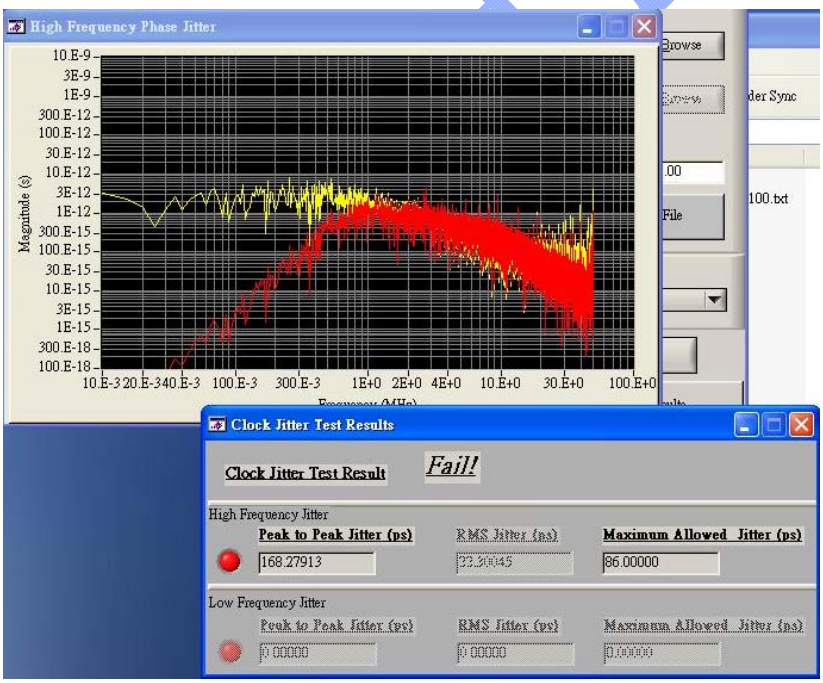
	Internal @-5℃
Period Jitter (ps) @10M Samples	-59.6~64.3 @P.10
Cycle-to-Cycle Jitter (ps) @10M Samples	-90.7~92 @P.10
Long Term Jitter (ps) Delay Time=-1uS] around 5 minutes	540
Long Term Jitter (ps) [Delay Time=-3uS] around 5 minutes	530
Long Term Jitter (ps) [Delay Time=-5uS] around 5 minutes	490
Long Term Jitter (ps) [Delay Time=-10uS] around 5 minutes	520 @P.11
Peak-to-Peak Phase Jitter (ps) by Clock Jitter Tool	194.27@P.12

p2p Phase Jitter @55 ℃ - Internal Clock:

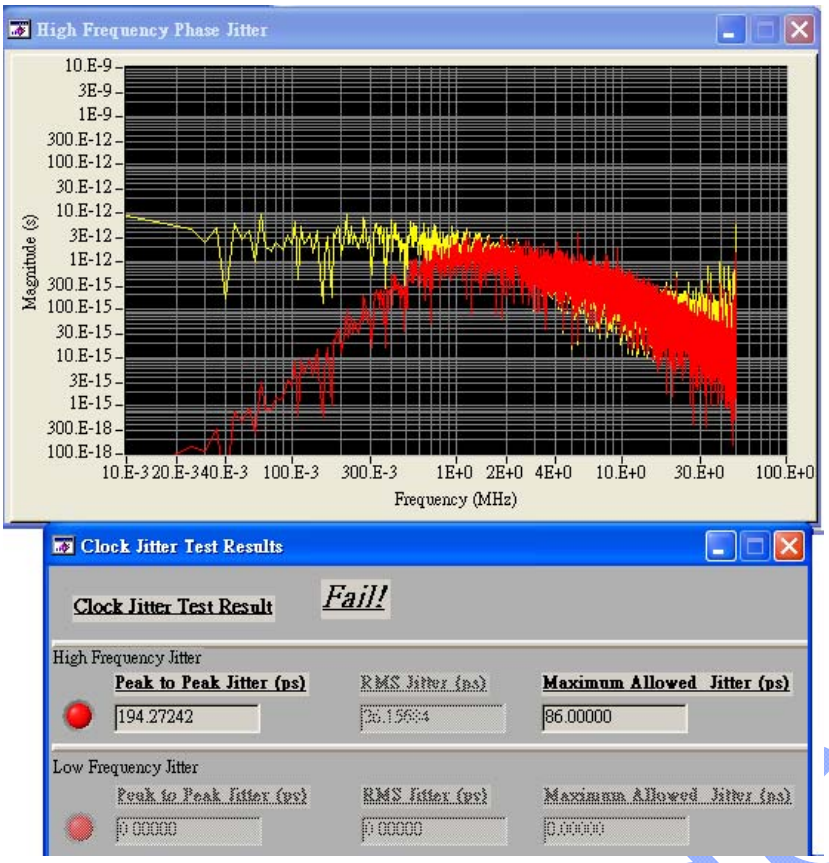
Confidential



p2p Phase Jitter @100 ℃ - Internal Clock:



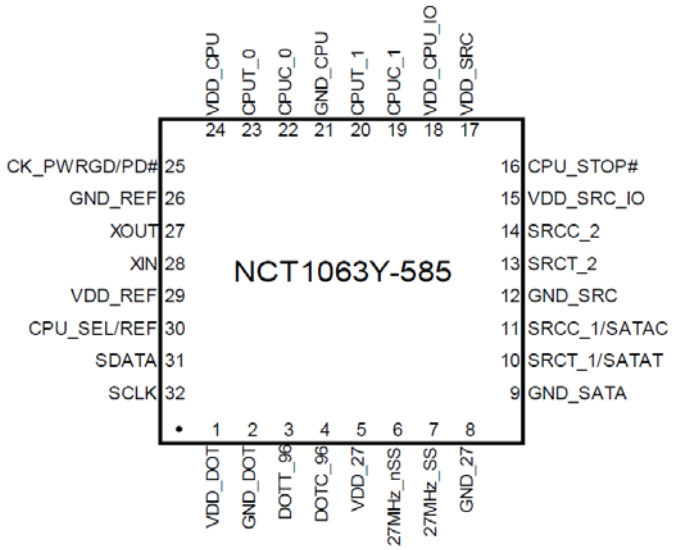
p2p Phase Jitter @-5 ℃ - Internal



1.23. PCIe External clock guideline

1.23.1 The external clock generator part number is **NCT1063Y-585** that we recommend. As follow pin out is specification of external clock generator:

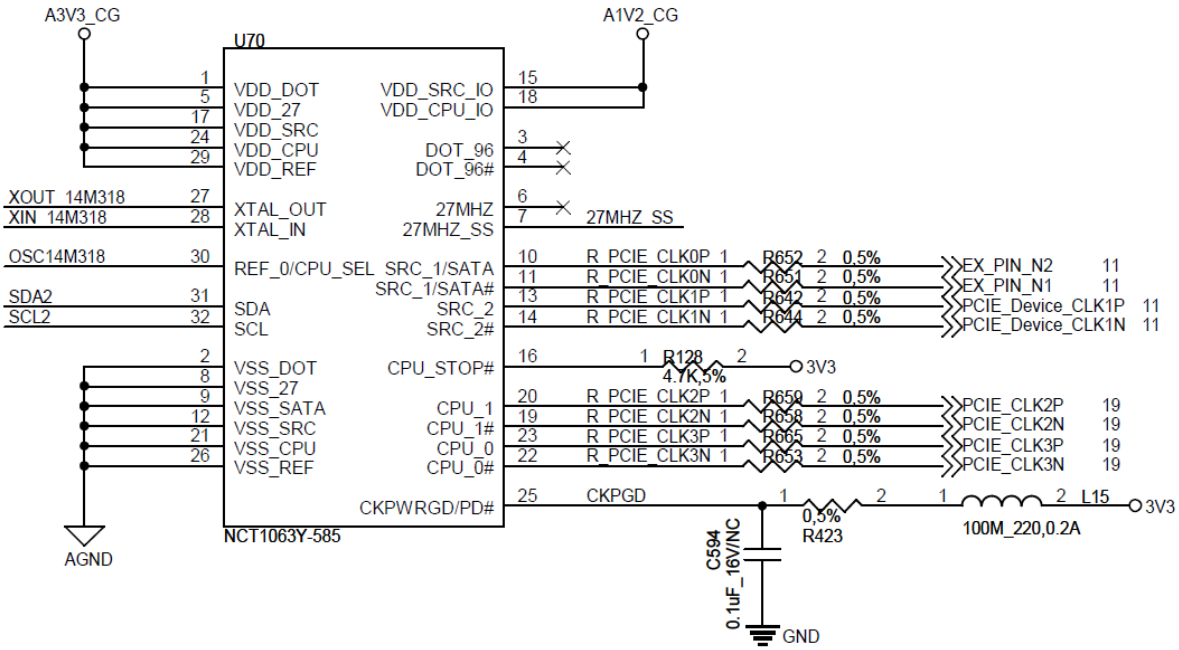
PIN CONFIGURATION



1.23.2 Differential pairs of Clock generator, pin of 10/11 connect to A9123 pin of N1 and N2 for PCIe internal

Confidential

PLL, and pin of 13/14 connect to PCIe device's differential input clock pair. The two trace width is 4.5 mil and space 5mil to get near 100ohm differential impedance, and keep minimum 30 mils space for other traces to reduce crosstalk as possible. The trace length of two differential signals must be match within 10mil. As follow circuit is specification of external clock generator that we recommend:



1.24. USB Port Power Source Notice

To avoid USB device power low drop cause system power jitter noise, tuning up to 5.2V for USB port power source is recommend.

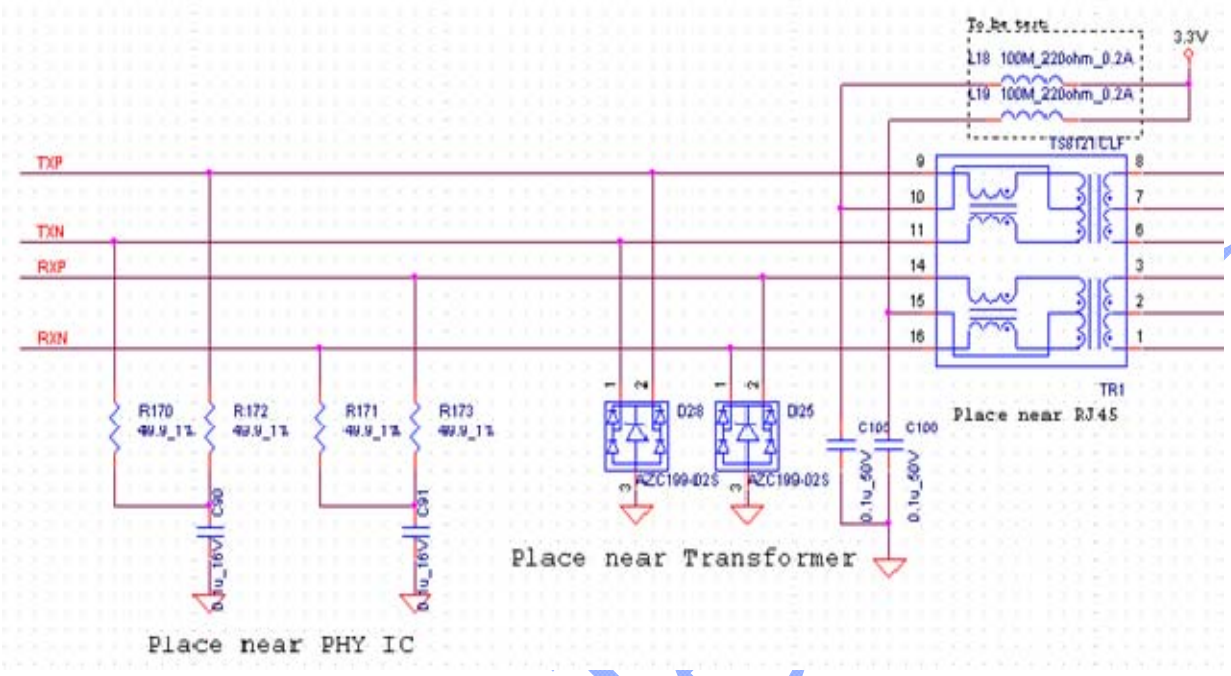
1.25. DDR3 support list

Vender	Density		
	128MB	256MB	512MB
NANYA	NT5CB128M8DN-DH	NT5CB256M8GN-CG	NT5CB512M8BN-CG
			NT5CC256M16CP-DII
			NT5CC256M16CP-DI
MICRON	MT41J128M8JP-15E		
	MT41J64M16JT-15E		

1.26. Ethernet protection scheme for POE (Power Over Ethernet) Device.

To avoid surge pulse cause Ethernet PHY damage, place TVS components to resist the surge pulse when connect with POE device, and TVS protect circuit have to use 2 high-speed I/O lines specification of TVS for each TXP/N and RXP/N differential pairs , can not use 4 high-speed I/O lines specification of TVS , both of TVS component place near by LAN

Transformer as following:



1.27. ADC Port Input Circuitry Protection Notice

. To avoid ADC input high pulse voltage made data read error from register , have to add as following circuit to protect ADC input normal function:

